

1995/96

1.2

$$\varphi = 30^\circ$$

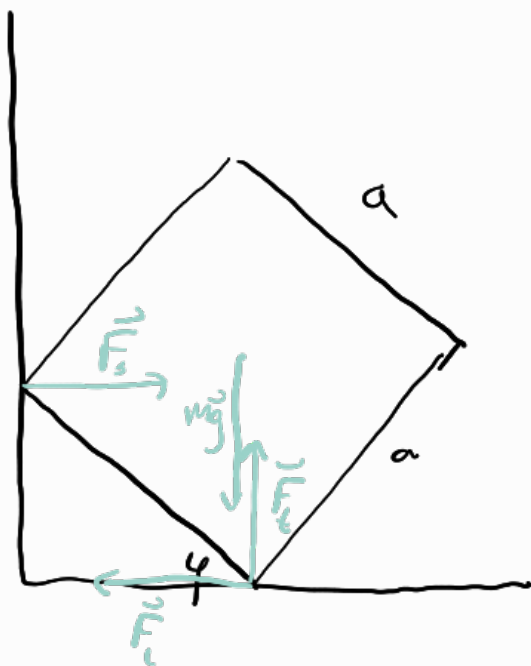
$$m = 12 \text{ kg}$$

$$a = 0,6 \text{ m}$$

$$k_L = 0,5$$

$$F_s = ?$$

$$k_{L\min} = ?$$



Side:

$$mg = F_t$$

$$F_s = F_L$$

Navori:

$$M_s = r F_s \sin \beta$$

$$M_t = r F_t \sin \alpha$$

$$M_l = -r F_l \sin \gamma$$

$$\alpha = \beta = 15^\circ$$

$$\gamma = 75^\circ$$

$$F_s \sin \beta + F_t \sin \alpha = F_l \sin \gamma$$

$$F_s = \frac{-F_g \sin \alpha}{\sin \beta - \sin \gamma} = 43 \text{ N}$$

za $k_{L\min}$:

$$F_L = mg k_{L\min}$$

$$k_{L\min} = \frac{43 \text{ N}}{120 \text{ N}} = 0,37$$

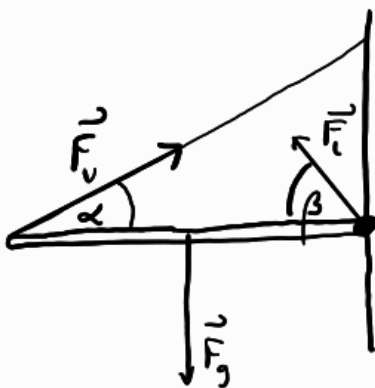
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$$F_g = 50 \text{ N}$$

$$\alpha = 30^\circ$$

$$F_v = ?$$

$$F_l = ?$$



$$F_g \cdot \frac{x}{2} = F_v \cdot x \cdot \sin \alpha$$

$$F_v = \frac{F_g}{2 \sin \alpha} = 50 \text{ N}$$

$$F_v \sin \alpha + F_l \sin \beta = F_g$$

$$F_v \cos \alpha = F_l \cos \beta$$

$$F_g = F_v (\sin \alpha + \cos \alpha \tan \beta)$$

$$\tan \beta = \frac{F_g - F_v \sin \alpha}{F_v \cos \alpha} = 0,577$$

$$\beta = 30^\circ$$

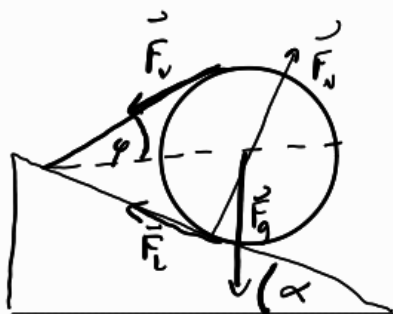
$$F_l = \frac{F_v \cos \alpha}{\cos \beta} = 50 \text{ N}$$

29/11

$$\alpha = 30^\circ$$

$$F_g = 50 \text{ N}$$

$$\varphi = 45^\circ$$



$$x: -F_l \cos \alpha - F_v \cos \varphi + F_N \sin \alpha = 0$$

$$y: F_l \sin \alpha - F_v \sin \varphi + F_N \cos \alpha - F_g = 0$$

navori:

$$F_l \cdot r \cdot \sin 90^\circ = F_v \cdot r \cdot \sin 90^\circ$$

$$F_l = F_v$$

$$F_v = F_l = \frac{F_g}{\sin \alpha - \sin \varphi + \frac{\cos \alpha + \cos \varphi}{\tan \alpha}} = 19,9 \text{ N}$$

$$F_l = F_N k_{\min}$$

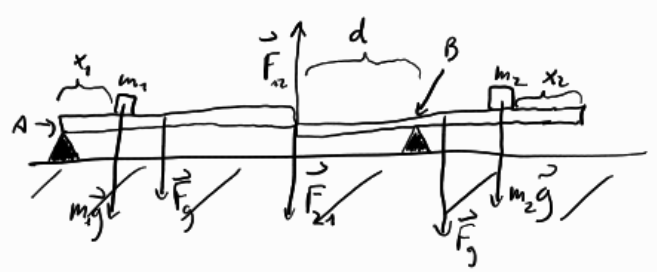
$$F_N = \frac{F_l (\cos \alpha + \cos \varphi)}{\sin \alpha} = 62,5 \text{ N}$$

$$k_{\min} = \frac{19,9}{62,5} = 0,32$$

$$F_l (\sin \alpha - \sin \varphi) + F_l \frac{\cos \alpha + \cos \varphi}{\tan \alpha} = F_g$$

29/8

$d = 1\text{m}$
 $x_1 = 1\text{m}$
 $l = 3\text{m}$
 $F_g = 150\text{N}$
 $m_1 = 80\text{kg}$
 $m_2 = 60\text{kg}$



$x_2 = ?$
 $v_2 = 10 \frac{\text{cm}}{\text{s}}$
 $v_1 = ?$

A:

$$x_1 m_1 g + \frac{l}{2} F_g - F_{z2} l = 0$$

$$F_{z2} = \frac{x_1 m_1 g + \frac{l}{2} F_g}{l} = 341,7\text{N} = F_{z1}$$

B:

$$\left(\frac{l}{2} - d\right) F_g + m_2 g (l - d - x_2) - d \cdot F_{z1} = 0$$

$$m_2 g x_2 = \left(\frac{l}{2} - d\right) F_g + m_2 g l - d m_2 g - d F_{z1}$$

$$x_2 = \frac{\left(\frac{l}{2} - d\right) F_g + m_2 g (l - d) - d F_{z1}}{m_2 g} = 1,56\text{m}$$

$$d \cdot F_{z1} = \frac{dx_1 m_1 g}{l} + \frac{d F_g}{2}$$

$$v_2 = \frac{dx_2}{dt} = - \frac{m_1 g \cdot d}{m_2 g l} \cdot \frac{dx_1}{dx_2} = - \frac{m_1}{m_2} \frac{d}{l} (v_1) \Rightarrow v_1 = v_2 \frac{m_2}{m_1} \cdot \frac{l}{d} = v_2 \cdot \frac{60}{80} \cdot \frac{3}{1} = \frac{9}{4} v_2 = 22,5 \frac{\text{cm}}{\text{s}}$$

29/30

$r = 25\text{cm}$
 $m = 1\text{kg}$
 $v_{z0} = 10\text{s}^{-1}$
 $k_{tr} = 0,2$
 $n_1, n_2 = ?$



$$J = I_1 = I_2$$

$$I_1 = J \omega_{z0}$$

$$I_2 = 2J \omega_k$$

$$\omega_{z0} = 2 \omega_k$$

$$\omega_k = \pi v_{z0}$$

$$J = m r^2$$

$$M_z = F_{tr} \cdot r = -J \alpha_z$$

$$-g k_{tr} r = -J \alpha_z$$

$$\alpha_z = -\frac{g k_{tr}}{r}$$

$$\alpha_1 = -\alpha_2$$

$$\omega_2 = \omega_{z0} + \alpha_2 t_k = \omega_{z0} - \frac{g k_{tr} t_k}{r}$$

$$\omega_1 = \alpha_1 t_k = \frac{g k_{tr} t_k}{r}$$

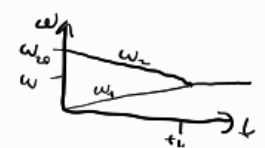
$$\omega_1 = \omega_2$$

$$2 \frac{g k_{tr} t_k}{r} = \omega_{z0}$$

$$t_k = \frac{r \pi v_{z0}}{g k_{tr}}$$

$$\varphi_1 = \frac{\alpha_1 t_k^2}{2} = \frac{\omega_{z0}^2 r}{8 g k_{tr}} = 2\pi \cdot 10$$

$N_1 = 10$



$$\varphi_2 = \omega_{z0} t_k + \frac{\alpha_2 t_k^2}{2}$$

$$= \omega_{z0}^2 \frac{r}{2 g k_{tr}} - \frac{g k_{tr} r \omega_{z0}^2}{2 g^2 k_{tr}^2}$$

$$= \frac{r \omega_{z0}^2}{2 g k_{tr}} \left(1 - \frac{1}{4}\right) = 2\pi \cdot 30$$

$N_2 = 30$

Is kan analog, ampak valji:

$$M_z = \int_0^R dF_{tr} \cdot r = g k_{tr} \int_0^R dm \cdot r = \frac{g k_{tr} m}{R^2} \int_0^R r^2 dr = \frac{2 g k_{tr} m R}{3} = -J \alpha_z$$

$$dm = \frac{2 \pi r dr}{R^2} m^2$$

$$J = \frac{1}{2} m R^2$$

$$\alpha_z = -\frac{4 g k_{tr}}{3 R}$$

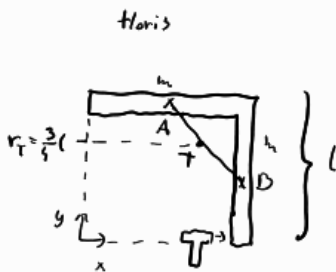
$$N_{zvalj} = 22,5$$

$$N_{1valj} = 7,5$$

2. kol.
90/01

3.

$$\frac{v_A}{v_B} = ?$$



$$K_A = \frac{m \frac{l}{2} + m l}{2 m} = \frac{3 l}{4} = r_T$$

$$G: x: \int F dt = 2 m v_T$$

$$\Gamma: \int M dt = J \omega_T = \int F dt r_T$$

$$J_A = J_B = \frac{1}{12} m l^2 + m \left(\frac{\sqrt{2}}{4} l \right)^2 =$$

$$d = \frac{\sqrt{2}}{4} l$$

$$= m l^2 \left(\frac{1}{12} + \frac{2}{16} \right) = \frac{5}{24} m l^2$$

$$J = 2 J_1 = \frac{5}{12} m l^2$$

$$2 m v_T \frac{3}{4} l = \frac{5}{12} m l^2 \omega_T$$

$$v_T = \frac{5 l \omega_T}{18}$$

$$v_{AT} = \omega_T \cdot \frac{\sqrt{2}}{4} l$$

$$v_{BT} = \omega_T \cdot \frac{\sqrt{2}}{4} l$$

$$\vec{v}_{AT} = \left(-\omega_T \frac{\sqrt{2}}{4} l \cdot \frac{\sqrt{2}}{4}, -\omega_T \frac{\sqrt{2}}{4} l \cdot \frac{\sqrt{2}}{4} \right) =$$

$$= -\frac{1}{4} \omega_T l (1, 1)$$

$$\vec{v}_{BT} = \frac{1}{4} \omega_T l (1, 1)$$

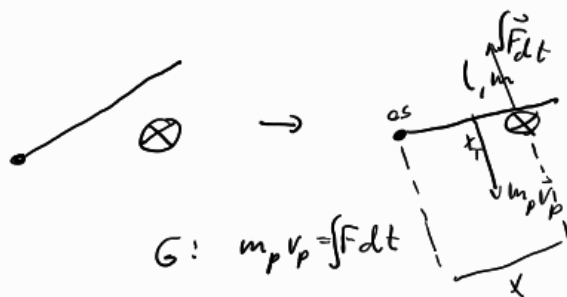
$$\vec{v}_A = \vec{v}_T + \vec{v}_{AT} = \left(\frac{1}{4} \omega_T l + \frac{5 l \omega_T}{18}, -\frac{1}{4} \omega_T l \right) = \omega_T l \left(\frac{1}{36}, -\frac{1}{4} \right)$$

$$\vec{v}_B = \vec{v}_T + \vec{v}_{BT} = \omega_T l \left(\frac{13}{36}, \frac{1}{4} \right)$$

$$\frac{v_A}{v_B} = \frac{\sqrt{\left(\frac{1}{36} \right)^2 + \left(\frac{1}{4} \right)^2}}{\sqrt{\left(\frac{13}{36} \right)^2 + \left(\frac{1}{4} \right)^2}} = \sqrt{\frac{41}{221}}$$

5.22 V stebel udarimo s kijem, po udarcu obmiruje

a) kje naj udarimo,
da v roki ne čutimo
sunka sile



$$G: m_p v_p = \int F dt$$

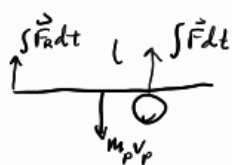
$$R: \int \omega = x \int F dt \quad J = \frac{1}{3} m l^2$$

$$\frac{1}{3} m l^2 \cdot \frac{v_p}{x} = x m_p v_p \quad \omega = v_p \cdot \frac{2}{l}$$

$$x = \frac{2}{3} l$$

b) $\int \vec{F}_2 dt$ v odv.

od razdalje



$$G: m_p v_p = \int \vec{F}_2 dt + \int \vec{F}_2 dt$$

$$R: \int \omega = x \int F_2 dt$$

$$\frac{1}{3} m l^2 \cdot \frac{v_p}{x} = x (m_p v_p - \int F_2 dt)$$

$$\int F_2 dt = \frac{m_p v_p (x - \frac{2}{3} l)}{x} = \underline{\underline{m_p v_p (1 - \frac{2}{3} \frac{l}{x})}}$$